```
!pip install -q ydata-profiling
```

```
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```

```
import pandas as pd
df = pd.read_csv('/content/WA_Fn-UseC_-Telco-Customer-Churn.csv')
```

```
from ydata_profiling import ProfileReport

profile = ProfileReport(
    df,
    title="Telecom Profile Report",
    explorative=True
)
```

```
/tmp/ipykernel_717/928863497.py:1: DeprecationWarning:
    `import ydata_profiling` is deprecated and will not receive more updates.
    Please install fg-data-profiling via `pip install fg-data-profiling` and use `import data_profiling` instead.

  from ydata_profiling import ProfileReport
```

```
profile.to_file("dataset_profile_report.html")
```

```
Summarize dataset: 100%    34/34 [00:06<00:00, 3.49it/s, Completed]

  0%|          | 0/21 [00:00<?, ?it/s]
  5%|▌         | 1/21 [00:00<00:04,  4.62it/s]
 24%|██▍       | 5/21 [00:00<00:01, 15.64it/s]
 38%|███▊      | 8/21 [00:00<00:00, 18.73it/s]
 52%|█████▏    | 11/21 [00:00<00:00, 18.34it/s]
 62%|██████▏   | 13/21 [00:00<00:00, 18.09it/s]
 71%|███████▏  | 15/21 [00:00<00:00, 17.71it/s]
 81%|████████  | 17/21 [00:01<00:00, 17.72it/s]
100%|██████████| 21/21 [00:01<00:00, 15.29it/s]
Generate report structure: 100%    1/1 [00:07<00:00, 7.23s/it]

Render HTML: 100%    1/1 [00:00<00:00, 1.59it/s]

Export report to file: 100%    1/1 [00:00<00:00, 37.13it/s]
```

```
contract_churn = pd.crosstab(
    df['Contract'],
    df['Churn'],
    normalize='index'
) * 100

print(contract_churn)
```

```
Churn                  No        Yes
Contract
Month-to-month  57.290323  42.709677
One year        88.730482  11.269518
Two year        97.168142   2.831858
```

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Start coding or generate with AI.

Start coding or generate with AI.

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```
import matplotlib.pyplot as plt

df.boxplot(column='tenure', by='Churn')
plt.title('Tenure vs Churn')
```